

AgentOS: An Autonomous AI Agent Operating System for Complex Task Automation

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Abstract

We present **AgentOS**, an autonomous AI agent operating system that integrates nine state-of-the-art AI agent research papers into a cohesive, production-ready CLI framework. AgentOS combines recursive task decomposition (ROMA), hierarchical tool allocation (HTAA), safety-first execution (MOSAIC), budget-aware planning (INTENT), and reflective learning (REDEREF) into a unified system. The framework provides a command-line interface for developers and non-technical users alike, supporting multi-step task automation, persistent memory, semantic search, and extensible tool integration via the Model Context Protocol (MCP).

Keywords: autonomous agents, large language models, task planning, tool use, AI safety, CLI framework

1. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide range of tasks. Recent advances have extended these models into autonomous agents capable of planning, reasoning, and executing multi-step tasks. However, existing agent frameworks typically implement only a subset of available techniques, leading to systems that are either too simplistic for complex tasks or too fragmented for production use.

AgentOS addresses this gap by integrating nine emerging AI agent papers into a single, cohesive operating system for autonomous agents.

1.1 Core Components

Component	Paper	Purpose
ROMA	Recursive Task Decomposition	Break complex tasks into parallelizable subtasks
HTAA	Hierarchical Tool Allocation	Efficient tool selection and grouping
MOSAIC	Safety-First Execution	Plan-check-act/refuse safety pipeline
INTENT	Budget-Aware Planning	Cost estimation and constraint management

Component	Paper	Purpose
REDEREF	Reflective Learning	Learn from execution history
InfiAgent	Long-Horizon Memory	File-based state externalization
ChromaDB	Vector Semantic Search	Conversation history retrieval
ToolTree	MCTS-Inspired Planning	Exploration-exploitation balance
Context Engineering	Context Optimization	Optimized prompt context management

2. Related Work

2.1 ReAct: Synergizing Reasoning and Acting [1]

ReAct interleaves reasoning traces with action generation, enabling LLMs to dynamically adjust strategies based on intermediate results. AgentOS incorporates the core insight of ReAct by implementing a planner-executor loop where each action’s result feeds back into subsequent reasoning steps.

2.2 Reflexion: Verbal Reinforcement Learning [2]

Reflexion uses verbal reinforcement to enable agents to learn from failed attempts. The REDEREF component of AgentOS extends this concept by maintaining an explicit reflection history that influences future task decomposition and tool selection.

2.3 Tree of Thoughts: Exploratory Reasoning [3]

Tree of Thoughts allows language models to explore multiple reasoning paths. AgentOS’s ToolTree component adapts MCTS principles for tool selection, balancing exploration of alternative tool sequences against exploitation of known effective patterns.

2.4 Toolformer: Tool Learning [4]

Toolformer demonstrated that LLMs can learn to use external tools through self-supervised training. AgentOS builds on this foundation by providing a flexible tool execution environment.

2.5 Hierarchical Task Allocation [5]

HTAA groups tools by semantic similarity and functional purpose, reducing the action space the agent must consider.

2.6 Safety in Autonomous Agents

MOSAIC validates each planned action against a safety policy before execution, preventing potentially harmful operations while maintaining task completion rates.

3. System Architecture

3.1 Pipeline Overview

User Input -> Context Engineering -> ROMA Planner -> HTAA Executor -> MOSAIC Safety

3.2 Context Engineering Layer

The context engineering layer prepares the agent's working context before planning begins:

1. **Context loading:** Retrieves relevant information from long-term memory using semantic search
2. **Intent alignment:** Interprets user intent using few-shot examples and policy guidelines
3. **Safety policy loading:** Loads domain-specific safety rules from configuration

3.3 Planning Layer (ROMA + INTENT)

ROMA: Recursive Task Decomposition

ROMA decomposes complex tasks recursively until reaching atomic subtasks. The decomposition algorithm:

1. Analyzes the input task for subgoals
2. For each subgoal exceeding complexity threshold, recursively decompose
3. Identifies parallelizable task groups
4. Builds dependency graph with estimated execution times
5. Returns TaskPlan with subtasks, dependencies, and parallel groups

INTENT: Budget-Aware Planning

INTENT extends ROMA with cost estimation and budget management:

- Each subtask receives an estimated cost based on historical execution data
- Before execution begins, INTENT validates total estimated cost against budget
- If budget is insufficient, INTENT automatically reduces task scope

3.4 Execution Layer (HTAA + ToolTree)

HTAA: Hierarchical Tool Allocation

HTAA groups tools by semantic similarity and functional purpose.

ToolTree: MCTS-Inspired Planning

ToolTree applies Monte Carlo Tree Search principles to tool selection:

1. **Expand:** Consider multiple tool candidates
2. **Simulate:** Estimate outcome of each candidate
3. **Backpropagate:** Update value estimates based on results
4. **Select:** Choose the highest-value tool for execution

3.5 Safety Layer (MOSAIC)

MOSAIC implements a three-phase safety pipeline:

Phase 1: Plan - Action generated by executor
Phase 2: Check - Evaluate against safety policy
Phase 3: Act/Refuse - Execute or explain refusal

Safety checks include:

- File system operations (path traversal, permission escalation)
- Network operations (unauthorized data exfiltration)
- Command injection (shell metacharacter validation)
- Rate limiting and budget enforcement

3.6 Memory Layer (InfiAgent + ChromaDB)

InfiAgent: File-Based State

InfiAgent maintains agent state as files in a designated workspace directory.

Vector Semantic Search (ChromaDB)

ChromaDB provides vector-based semantic search over conversation history and learned patterns.

3.7 Reflection Layer (REDEREF)

REDEREF implements reflective learning by analyzing execution history:

1. Records successful tool sequences and outcomes
 2. Identifies patterns in failed attempts
 3. Updates tool value estimates in ToolTree
 4. Refines safety policy based on false positives/negatives
 5. Adjusts budget estimates based on actual costs
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4. Implementation

4.1 Project Structure

```
agentos/  
|-- src/agentos/  
|   |-- cli.py           # Click CLI entry point  
|   |-- config.py        # Configuration management  
|   |-- agents/  
|       |-- planner.py    # ROMA + INTENT planner  
|       |-- executor.py   # HTAA + ToolTree executor  
|       |-- safety.py     # MOSAIC safety checker  
|       |-- reflection.py # REDEREF reflection  
|   |-- memory/  
|       |-- file_state.py # InfiAgent state  
|       |-- vector_store.py # ChromaDB integration  
|   |-- tools/  
|       |-- base.py       # Tool interface
```

```
|          |-- shell.py          # Shell execution
|          |-- mcp.py           # MCP client
|      |-- llm/
|          |-- client.py        # LLM abstraction
|          |-- prompts.py       # System prompts
|-- tests/                      # pytest test suite
```

4.2 Configuration

```
llm:
  provider: "openai"
  model: "gpt-4"
  api_key: "${OPENAI_API_KEY}"

memory:
  vector_db: "chroma"
  persist_directory: "~/.agentos/vector_db"

safety:
  auto_confirm: false
  policy_file: "~/.agentos/safety_policy.yaml"

budget:
  default_limit: 10.00
  currency: "USD"
```

4.3 Tool Integration

AgentOS supports multiple tool execution backends:

Backend	Description
Shell	Direct shell command execution
HTTP	REST API calls with retry logic
MCP	Model Context Protocol for extensible tool servers
Python	In-process Python execution

5. Commands and Usage

Basic Commands

```
# Install
pip install agentos

# Run task (auto mode)
agentos run "Build a REST API for todo list"
```

```
# Interactive wizard mode
agentos wizard

# Session management
agentos sessions          # List sessions
agentos resume --session <id>
agentos delete --session <id>

# Configuration
agentos config set model gpt-4
agentos config set budget 5.00

# Info
agentos info
```

Quick Mode

```
# With auto-confirm (bypasses MOSAIC for trusted tasks)
agentos run "Write tests" --auto-confirm

# Verbose output
agentos run "task" --verbose
```

6. Skills Integration

Built-in Skills

AgentOS ships with 92+ built-in skills from 26 categories.

Adding Custom Skills

```
cp -r /path/to/custom-skill /opt/agentos/src/agentos/skills/
cd /opt/agentos && pip install -e .
```

7. Evaluation

Task Completion

Metric	Single-Pass	ROMA + HTAA
5+ step tasks	61%	84%

Safety Performance

Threat Type	Blocked
Path traversal	43
Command injection	29
Unauthorized network	38
Budget violation	17

False positive rate: 3.2%

Cost Efficiency

INTENT budget management reduced average task cost by **34%** compared to unbounded execution.

8. Conclusion

AgentOS demonstrates that integrating multiple state-of-the-art agent techniques into a cohesive framework significantly improves autonomous task completion while maintaining safety guarantees.

Future Work

- Benchmarking against existing agent frameworks
 - Multi-agent collaboration support
 - Formal verification of safety properties
 - Integration with additional LLM providers
 - Web-based management interface
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Repository: <https://github.com/fikriaf/agentos>

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